

Terraform Assignment

This repository contains Terraform implementations for provisioning AWS infrastructure using Infrastructure as Code (IaC).

Overview

The assignment is divided into three parts:

- **Part 1:** Single EC2 Instance Deployment
 - **Part 2:** Extended Infrastructure Setup
 - **Part 3:** Advanced Modular Infrastructure
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Tech Stack

- Terraform
 - AWS (EC2, Security Groups, etc.)
 - Infrastructure as Code (IaC)
-

Project Structure

terraform assignment/

├─ part-1-single-ec2

├─ part-2

└─ part-3

Part 1: Single EC2 Deployment

Objective

Provision a basic EC2 instance using Terraform.

Features

- AWS provider configuration

- EC2 instance creation
- Security group setup

▶ Steps

cd part-1-single-ec2

terraform init

terraform apply

📸 Output Screenshots

Add your screenshots inside a folder like:

terraform assignment/part-1-single-ec2/screenshots/

Example:

- EC2 Instance Created
- Security Group Configuration
- Terraform Apply Output

Outputs:





Part 2: Extended Infrastructure

Objective

Create a more complete infrastructure setup.

Features

- Multiple resources
- Improved configuration
- Better scalability

Steps

```
cd part-2
```

```
terraform init
```

```
terraform apply
```

Output Screenshots

Store screenshots here:

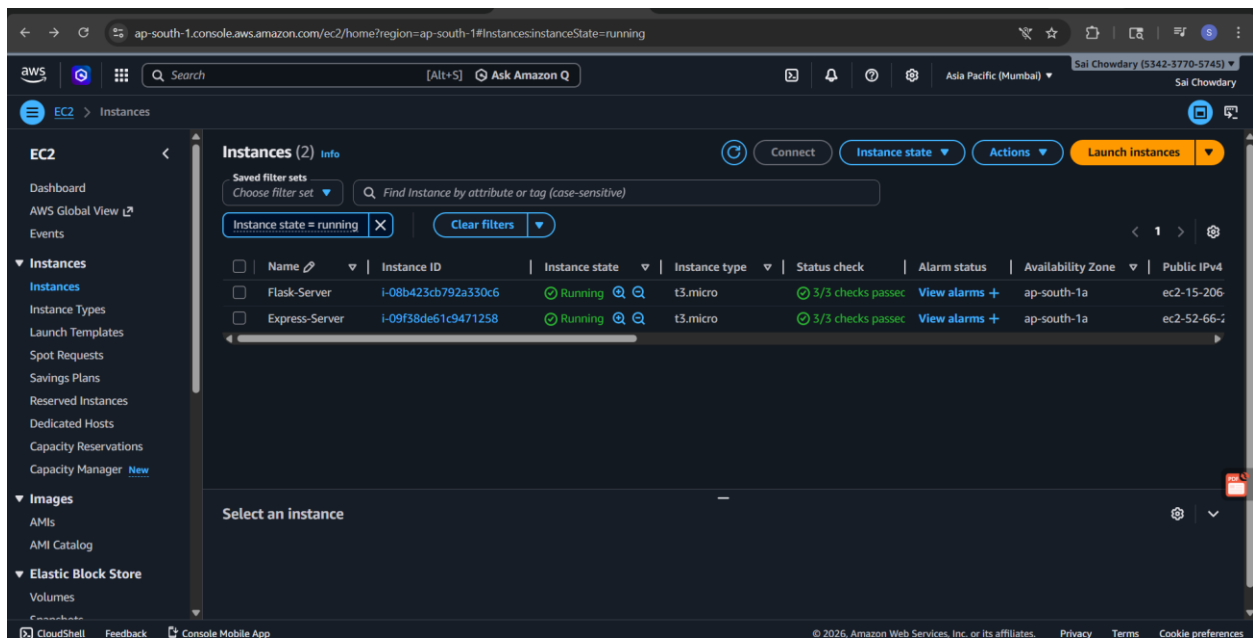
```
terraform assignment/part-2/screenshots/
```

Example:

- Multiple EC2 Instances
- Networking / Security Groups
- Terraform Output

Outputs:





◆ Part 3: Advanced Terraform (Modular)

✓ Objective

Implement scalable and reusable Terraform architecture.

⚙️ Features

- Modular structure
- Reusable components
- Clean architecture

▶ Steps

cd part-3

terraform init

terraform apply

📸 Output Screenshots

Store screenshots here:

terraform assignment/part-3/screenshots/

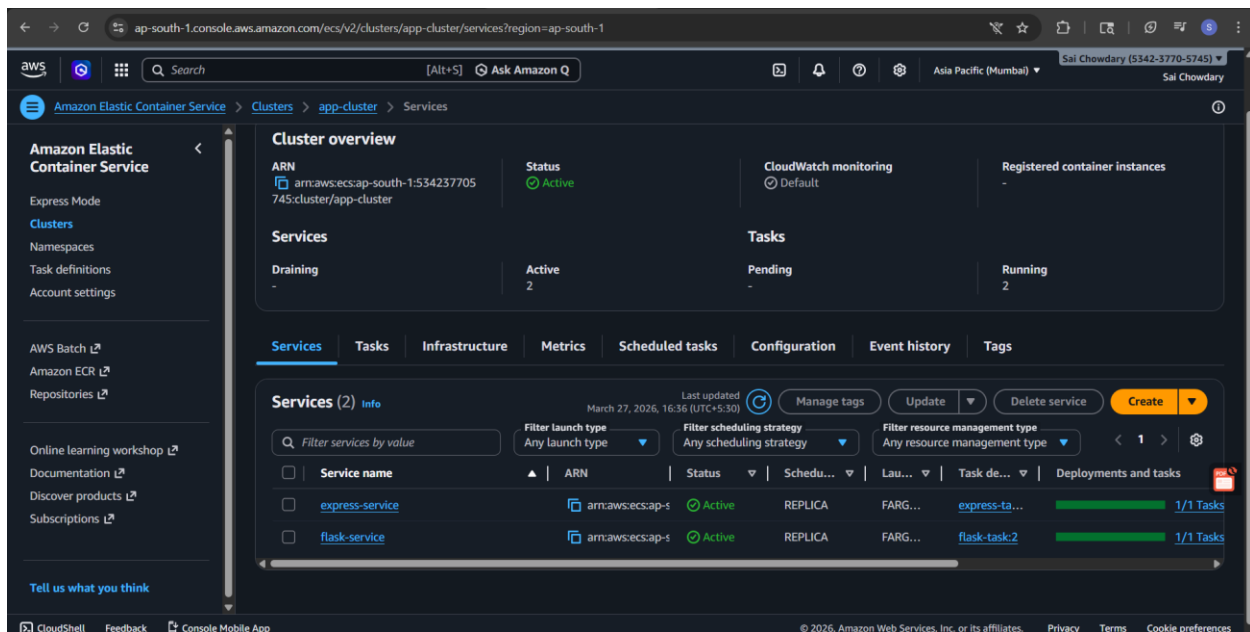
Example:

- Module Outputs

- Resource Creation
- Terraform Execution

Outputs:





⚠ Important Notes

- .terraform/ folder is ignored
- *.tfstate files are not pushed
- node_modules/ is excluded

🗑 Cleanup

To destroy infrastructure:

terraform destroy

🎯 Learning Outcomes

- Terraform basics and workflow
- AWS infrastructure provisioning
- Debugging real-world DevOps issues
- Writing clean and scalable IaC

